

Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru-3bb728174 | github.com/zac-09 | Kampala, Uganda (UTC+3, within CET±2)

PROFESSIONAL SUMMARY

Product engineer with 6 years owning features end-to-end — from problem definition to production. Core stack: TypeScript, React, and Node.js, with Python for AI backends. Built conversational AI in production (WhatsApp healthcare chatbot: OpenAI + Google Vision prescription OCR) and an open-source agentic pipeline on Claude Code. At MTailor (US e-commerce startup, reporting to the CTO), led a zero-downtime migration of 20+ production apps to Firebase/GCP with a real-time two-way Pub/Sub sync, and shipped revenue features: Express Shipping (+\$40/order) and a 3D visualization that lifted buyer conversion. Startup-tested at every stop — thin specs, fast shipping, measurable outcomes. Kampala, UTC+3: within CET±2 of Berlin year-round.

CORE COMPETENCIES

End-to-End Product Ownership

TypeScript · React · Node.js

AI & Agent Systems (Python, OpenAI)

Real-Time Data Systems

Scalable Systems Architecture

Startup Velocity

WORK EXPERIENCE

MTailor (US Remote) Sep 2022 – Present

Software Engineer — reporting directly to the CTO

- Shipped **Express Shipping end-to-end** (concept → pricing → backend → ops handoff), generating an additional **\$40 revenue per order**
- Built a **3D visualisation feature** for customers using video overlay and ffmpeg, **increasing buyer conversion**
- Led the zero-downtime migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP
- Implemented real-time two-way sync between MongoDB and Firestore using Node.js and Google Pub/Sub
- Migrated file storage from Amazon S3 to GCS via a Python script; moved all services off AWS, saving **\$5,000/month**
- Migrated a Webflow website from a Parse backend to Firebase

MTailor (US Remote) Mar 2022 – Jul 2022

Contractor (Backend Migration)

- Hired via Upwork onto an unfamiliar stack; built MongoDB→Firestore migration scripts with complex processing logic
- Wrote Firebase SDK documentation (Node.js and Python) other engineers onboarded from; trained the Ops team — converted to full-time and given the company-wide migration lead within months

CodeBits (Uganda) Jan 2020 – Jul 2021

Led a team of 4 developers building legal tech for NGOs end-to-end

- Owned client relationships and delivery for FIDA Uganda and LASPNET — requirements, architecture, build, ship
- Architected an event-driven microservices backend using **Apache Kafka, Docker, and Kubernetes**
- Built a **USSD service** for legal aid over feature phones, communicating with a Node.js backend via gRPC — meeting rural users on the infrastructure they actually have
- Developed a cross-platform mobile app using **React Native and Expo** (iOS & Android) with Firebase push notifications

Dr Wealth (Singapore Remote) Aug 2021 – Jan 2022

Software Engineer

- Led optimisation of a PWA delivering real-time stock prices; built jobs querying 2M+ Firestore records from Morningstar APIs
- Extended backend APIs with Firebase Cloud Functions and Express.js; built responsive React/Tailwind UI pages

Mind2matter (US Remote) Jan 2022 – Jul 2022

Software Engineer (Full-Stack)

- Built backends for DeFi applications using Web3 and Node.js; delivered pixel-accurate React UIs from Figma under tight timelines

PROJECTS

Healthcare WhatsApp Chatbot (conversational AI in production)

Conversational AI over the WhatsApp Business API: integrated OpenAI + Google Vision for prescription OCR — dialogue flows, third-party platform constraints, production operation.

Node.js · Python · OpenAI · Google Vision · WhatsApp Business API

career-ops — Agentic Pipeline (open source)

Agentic system built on Claude Code: Playwright ground-truth verification, batch evaluation workers, automated document generation. 100+ evaluations run.

Claude Code · Playwright · Node.js · YAML

Marketplace Recommendation Engine

Collaborative + content-based filtering engine in MongoDB for service-provider matching in a live marketplace.

Node.js · MongoDB

EDUCATION

BSc Computer Engineering — [Makerere University](#), Kampala, Uganda

2018 – 2022

SKILLS

Proficient: Node.js, TypeScript/JavaScript, React, Firebase, GCP, Python, MongoDB, SQL, Git, Linux

Intermediate: Go, AWS, Rust, React Native **Infrastructure:** Kubernetes, Docker, Kafka, gRPC, Pub/Sub **Languages:** English, Kiswahili